

15/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

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AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

CC-115

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MM : 720

CST-5

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Biology, Chemistry and Physics having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



BIOLOGY

- Property of a tissue is
 - (1) Same as that of its constituent cells
 - (2) A result of interactions among the constituent cells
 - (3) Similar to the properties of cell organelles of constituent cells
 - (4) A result of interaction among the molecular component comprising the organelles
- All viruses are/have
 - (1) Protein coat and DNA
 - (2) Parasites
 - (3) Both DNA and RNA
 - (4) Inert inside living cell
- "Kingdom Protista forms the link with kingdom plantae, animalia and fungi".
Select the option which is **incorrectly** matched w.r.t. slime moulds and their plant, animal or fungi like features.
 - (1) Naked plasmodium - Animal like feature
 - (2) Spore with true wall - Plant like feature
 - (3) Formation of fruiting body - Fungi like feature
 - (4) Formation of plasmodium - Protozoan like feature
- Identify the organism which causes white spots seen on mustard leaves and select the statement not true for it.
 - (1) Dikaryophase is the dominant phase of its life
 - (2) Asexually reproduce by spores
 - (3) Cell wall is made up of chitin and polysaccharides
 - (4) It produces sexual spores
- Match the following columns and select the **correct** option regarding bryophytes.

(a)	Gametophyte	(i)	Haploid, antherozoid producing structure
(b)	Sporophyte	(ii)	Flask shaped haploid structure
(c)	Antheridium	(iii)	Produces gametes
(d)	Archegonium	(iv)	Non photosynthetic diploid body

 - (1) a(iii), b(iv), c(ii), d(i)
 - (2) a(ii), b(i), c(iii), d(iv)
 - (3) a(iii), b(iv), c(i), d(ii)
 - (4) a(iv), b(iii), c(i), d(ii)
- The protoplast of brown alga contains all, **except**
 - (1) Plastids
 - (2) Centrally located vacuole
 - (3) Nucleus
 - (4) Pyrenoids
- Select the plant that has half-inferior ovary with apocarpous condition.
 - (1) Sunflower
 - (2) Mustard
 - (3) Rose
 - (4) Guava
- Commonly, the plant product used to arrest cell division can be isolated from the family
 - (1) Solanaceae
 - (2) Fabaceae
 - (3) Brassicaceae
 - (4) Liliaceae
- Select the **incorrect** statement.
 - (1) The cells of elongation zone of root gradually differentiate & mature
 - (2) The root hairs absorb water and minerals from soil
 - (3) One of the functions of root is synthesis of plant growth regulators
 - (4) Meristematic cells are thick walled with dense protoplasm
- Match the column-I with column-II and select the **correct** option.

Column-I	Column-II
a. Staminate	(i) Petals united
b. Syncarpous	(ii) Sterile stamen
c. Gamopetalous	(iii) Sepals free
d. Polysepalous	(iv) Fused carpels

 - (1) a(iv), b(iii), c(ii), d(i)
 - (2) a(iii), b(ii), c(iv), d(i)
 - (3) a(ii), b(iv), c(i), d(iii)
 - (4) a(i), b(ii), c(iii), d(iv)

11. Read the following assertion (A) and reason (R) statements and select the **correct** option
Assertion (A) : Intercalary meristem occurs between mature tissues.
Reason (R) : Intercalary meristem helps in the elongation of the organ.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)
 - (3) (A) is true but (R) is false
 - (4) Both (A) and (R) are false
12. Read the given statements and identify the correct one(s).
A. Mesophyll cells are devoid of chlorophyll and do not perform photosynthesis.
B. In monocot stem, the peripheral vascular bundle are generally smaller than the centrally located ones.
C. Dorsiventral leaf vascular bundles are surrounded by layer of thick walled bundle sheath cells.
D. The monocot stem has parenchymatous hypodermis.
- (1) B only
 - (2) B and C only
 - (3) All A, B, C & D
 - (4) C only
13. Outer side of epidermis is often covered by waxy material called a which is absent in b.
Select the correct option to fill in the blank a and b respectively.
- (1) Cuticle; roots
 - (2) Suberin; stems
 - (3) Cuticle; stems
 - (4) Suberin; roots
14. In the structure of plasma membrane, it is seen that hydrophobic tails are arranged towards inner side. This ensures that
- (1) Polar heads can be protected from aqueous environment
 - (2) Polar molecules can be transported through lipid bilayer
 - (3) Non polar tail of saturated hydrocarbons is protected from the aqueous environment
 - (4) Lipid soluble molecules can pass through tunnel proteins
15. RER is frequently observed in the cells actively involved in protein synthesis and secretion. The protein synthesized by ribosomes on RER are
- a. Lysosomal enzymes
 - b. Proteins destined for the ER
 - c. Proteins destined for the golgi
 - d. Proteins of cell membrane
 - e. Protein to be secreted outside the cell
- (1) Only a and b
 - (2) Only b and c
 - (3) Only a, b and d
 - (4) All a, b, c d and e
16. Division of labour is not exhibited by
- (1) *Paramecium*
 - (2) *Hydra*
 - (3) *Taenia*
 - (4) *Periplaneta*
17. Select an acidic amino acid among the following.
- (1) Palmitic acid
 - (2) Glutamic acid
 - (3) Uridylic acid
 - (4) Thymidyllic acid
18. Simple epithelium is present in/at
- (a) Diffusion boundary of alveoli
 - (b) Filtration membrane inside renal corpuscles
 - (c) Germinal epithelium of testis and ovary
 - (d) Inner lining of parts of nephrons and pancreatic ducts
- Choose the **correct** option.
- (1) (a) and (b) only
 - (2) (b) and (c) only
 - (3) (a), (b) and (d)
 - (4) (a), (b) and (c)
19. The skin serves as a respiratory organ in all of the following animals, **except**
- (1) Earthworm
 - (2) Frog
 - (3) Toad
 - (4) Cockroach
20. Stimulation of sympathetic nervous system has the same effect on heart beat rate as that of release of
- (a) Acetylcholine
 - (b) Adrenaline
 - (c) Noradrenaline
- Choose the correct option.
- (1) Only (a) and (b)
 - (2) Only (c)
 - (3) Only (b) and (c)
 - (4) (a), (b) and (c)

21. The most common type of bond present in both primary and tertiary structures of a protein is

- (1) Hydrogen bond
- (2) Disulphide bond
- (3) Ionic bond
- (4) Peptide bond

22. The presence of disulphide bond can be observed in all of the following compounds, **except**

- (1) Insulin
- (2) Pro-insulin
- (3) Antibody
- (4) Glycogen

23. **Assertion (A):** In lower invertebrates like sponges, coelenterates and flatworms, diffusion of gases occurs through their body surfaces.

Reason (R): Mechanism of breathing varies among different groups of animals depending mainly on their habitats and levels of organisation.

In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true statement but (R) is false
- (4) Both (A) and (R) are false statements

24. α -amino acids are substituted methanes. There are 20 types of amino acids present in proteins and all of them contain a single asymmetric carbon, **except**

- (1) Glycine
- (2) Proline
- (3) Lysine
- (4) Cysteine

25. The value of $p\text{CO}_2$ in alveolar air is

- (1) Equal to the value of $p\text{CO}_2$ in deoxygenated blood of systemic veins
- (2) More than the value of $p\text{O}_2$ in oxygenated blood of systemic artery
- (3) Less than the value of $p\text{O}_2$ in oxygenated blood of pulmonary vein
- (4) Equal to the value of $p\text{CO}_2$ in deoxygenated blood of pulmonary artery

26. Chlorophyll pigments are present in

- (1) Stroma
- (2) Thylakoids
- (3) Matrix
- (4) Lysosome

27. The most dramatic period of the cell cycle does **not** involve

- (1) Duplication of centrioles
- (2) Disintegration of nuclear envelope
- (3) Decondensation of chromosomes
- (4) Development of nuclear envelope

28. Spindle fibres begin to form during (i) and attach to specific site of chromosomes during (ii).

Select the **correct** option to fill in the blanks (i) and (ii).

	(i)	(ii)
(1)	Prophase	Metaphase
(2)	Anaphase	Telophase
(3)	Metaphase	Anaphase
(4)	Prophase	Anaphase

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

29. Which of the following sequences regarding different phases of a cell is acceptable?

- (1) $M \rightarrow G_1 \rightarrow G_0 \rightarrow G_1$
- (2) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$
- (3) $M \rightarrow S \rightarrow G_1 \rightarrow G_0$
- (4) $G_0 \rightarrow G_1 \rightarrow G_2 \rightarrow S$

30. State the following statement as True (T) or False (F) and select the correct option.

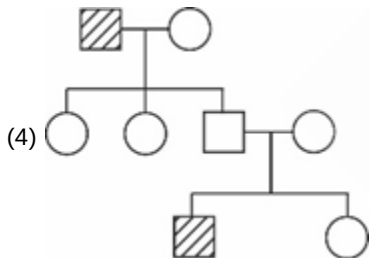
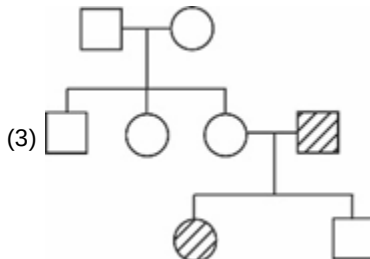
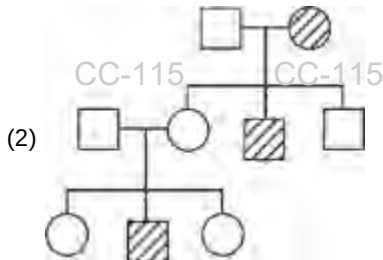
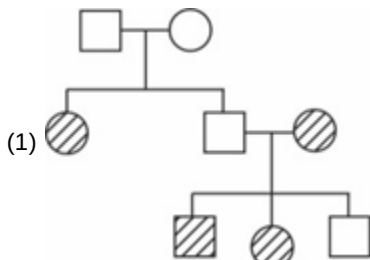
- (a) NADP reductase is located on the lumen side of the thylakoid membrane.
- (b) Reduction step of C_3 cycle involves utilization of three molecules of ATP for reduction and two of NADPH for phosphorylation per CO_2 molecule fixed.
- (c) Out of three, two products of light reactions are utilized in biosynthetic phase, while one diffuses out.
- (d) Cyclic photophosphorylation occurs when light of wavelengths below 680 nm are available for excitation.

	a	b	c	d
(1)	T	F	T	F
(2)	F	F	T	F
(3)	T	T	F	T
(4)	F	F	F	T

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

31. Metaphase I is different from metaphase II as in the former
- (1) Bivalent chromosomes align on the equatorial plate
 - (2) Homologous chromosomes separate from each other
 - (3) Crossing over between non-sister chromatids of homologous chromosomes occur
 - (4) Metaphasic plate is not formed
32. The proton gradient across the mitochondrial membrane is a result of
- (1) Energy of oxidation - reduction
 - (2) Chemiosmosis
 - (3) Flow of H^+ via F_0 particle
 - (4) Production of GTP in mitochondrial matrix
33. In the equation given below 'X', 'Y' and 'Z' respectively represents,
- $$\text{Pyruvic acid} + X + NAD^+ \xrightarrow{Y} Z + CO_2 + NADH + H^+$$
- (1) O_2 , Zn^{2+} , oxaloacetic acid
 - (2) CoA, Mg^{2+} , acetyl CoA
 - (3) CoA, Zn^{2+} , succinyl CoA
 - (4) O_2 , Mg^{2+} , succinyl-CoA
34. Among the following substances, select the substance/(s) that can be employed to delay leaf senescence.
- a. Autoclaved herring sperm DNA
 - b. Corn-kernel suspension
 - c. Coconut milk
- (1) Only a
 - (2) Only a and c
 - (3) Only b and c
 - (4) All a, b and c
35. A PGR which is isolated from coconut milk
- (1) Promotes cell division
 - (2) Increases the yield of sugarcane by nearly 20 tonnes per acre
 - (3) Is used to prepare weed-free lawns by gardener
 - (4) Is widely applied in hedge-making
36. Read the following assertion (A) and reason (R) statements and select the **correct** option
- Assertion (A)** : Embryo development precedes endosperm development.
- Reason(R)** : Early stages of embryogeny are different in both monocots and dicots though seeds are similar.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)
 - (3) (A) is true but (R) is false
 - (4) Both (A) and (R) are false
37. The pollen grains in the angiosperms represent the
- (1) Male gametes
 - (2) Microsporangium
 - (3) Megaspore
 - (4) Male gametophyte
38. α Thalassemia is controlled by two closely linked genes HBA1 and HBA2. In a man suffering from this disorder both the genes are mutated, but during the process of spermatogenesis normal condition in one of the genes HBA2 is restored. If this sperm fertilises an egg having mutated HBA1 gene then which of the following statements regarding the resulting child would be true?
- (1) This child either male or female will not show any symptom of the trait
 - (2) The child will show the trait only if he is a male
 - (3) This child will show mild symptom of the trait regardless of the gender
 - (4) The child will show the trait only if she is a female
39. Recombination frequency
- (1) Is directly proportional to the distance between the genes
 - (2) Is not useful to prepare the genetic maps
 - (3) Of closely situated genes is very high
 - (4) Is directly proportional to linkage

40. Which amongst the following pedigree charts best represents the inheritance of traits such as Myotonic dystrophy?
[Here, shaded symbols represent the affected ones.]



41. Choose the correct option to complete the analogy w.r.t an adult human.
Acetabulum : Coxal bone :: _____ : Scapula

- (1) Glenoid cavity
- (2) Hyoid
- (3) Ilium
- (4) Sternum

42. In athletes, all of the following observations are seen compared to a normal person during resting conditions, **except**

- (1) Decrease in heart rate
- (2) Heart muscles become stronger through regular exercise
- (3) Increase in stroke volume
- (4) Decrease in cardiac output

43. To complete the circuit of current flow during impulse conduction through a nerve fibre, the current

- (1) Always flows towards axon terminals on outer surface of axonal membrane.
- (2) Always flows towards cyton on the inner surface of axonal membrane.
- (3) Flows towards axon terminal on inner surface and towards cyton on outer surface of axonal membrane.
- (4) Always flows towards axon terminal on both inner and outer surfaces of axonal membrane.

44. **Assertion (A)** : Na^+ reabsorption from glomerular filtrate can occur in PCT, ascending limb of loop of Henle and collecting duct.

Reason (R) : Na^+ is always reabsorbed by passive process in PCT, ascending limb of loop of Henle and collecting duct. In the light of above statements, choose the correct option

- (1) Both (A) and (R) are true and (R) is the correct explanation of the (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)
- (3) (A) is true statement but (R) is false
- (4) Both (A) and (R) are false statements

45. Which of the following hormones is secreted by an endocrine structure that is situated outside the cranium?

- (1) Melatonin
- (2) Vasopressin
- (3) Thymosin
- (4) Oxytocin

46. Select the diploblastic, radially symmetrical animal from the options given below.

- (1) Sea anemone
- (2) Tapeworm
- (3) Hookworm
- (4) Pearl oyster

47. The organs which are the site for information processing and control in humans are

- a. Brain
- b. Spinal cord
- c. Muscles
- d. Blood

Choose the **correct** option.

- (1) a and d
- (2) b and c
- (3) a and b
- (4) a, b, c and d

48. The Medical Termination of Pregnancy (Amendment) Act, 2017 was enacted by the Government of India with all of the following intentions, **except**

- (1) Reducing the incidence of illegal abortion
- (2) Reducing maternal mortality
- (3) Reducing maternal morbidity
- (4) Increasing the incidence of female foeticide

49. Choose the **incorrect** match from the following w.r.t reptiles.

(1)	<i>Brachiosaurus</i>	–	Long-necked herbivore
(2)	<i>Stegosaurus</i>	–	Plated-back herbivore
(3)	<i>Pteranodon</i>	–	Flying reptile
(4)	<i>Triceratops</i>	–	Bipedal carnivore

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

50. A student labelled a male frog as sample 'A' and a female frog as sample 'B'. He dissected the samples and cut their ureters to make notable differences between both sexes. Select the option that correctly represents the correct difference between them.

Sample A

Sample B

- | | |
|---|--|
| (1) Bidder's canal absent | Bidder's canal present |
| (2) Gamete transport affected | No effect on gamete transport |
| (3) No functional connection between kidneys and gonads | Functional connection present between kidneys and gonads |
| (4) Fat bodies present | Fat bodies absent |

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

51. How many of the factors mentioned in the box below are not **true** for Darwinian theory of evolution?

Directional variation, Saltation, Minor variation, Inheritable 'factors', Heritable variations, Reproductive fitness

Select the **correct** option.

- (1) Four
- (2) Two
- (3) Three
- (4) Five

52. Hormone released from _____ facilitates water reabsorption from latter parts of the tubule and also affects the kidney functions by its constrictory effect on blood vessels.

Select the **correct** option to fill the blank.

- (1) Adrenal medulla
- (2) Anterior pituitary
- (3) Testes
- (4) Neurohypophysis

53. A 10 kb DNA molecule having two restriction sites for *EcoRI* was completely digested using *EcoRI* and the result obtained after agarose gel electrophoresis is shown below. Lane 1 – Loaded with the undigested DNA sample that will act as a control. Lane 2 – Loaded with the digested DNA samples



Choose the option which holds true for the above mentioned condition.

- (1) The DNA sample in lane 2 is a 10 kb plasmid.
- (2) The smallest fragment of DNA in lane 2 is closer to cathode.
- (3) The largest fragment in lane 2 is of 10 kb size.
- (4) The undigested DNA in lane 1 is a 10 kb linear DNA.

54. Match column I with column II and select the **correct** option.

Column I		Column II	
a.	Thomas Malthus	(i)	Malay Archipelago
b.	Louis Pasteur	(ii)	H.M.S. Beagle
c.	Ernst Haeckel	(iii)	His work on populations influenced Darwin
d.	Alfred Wallace	(iv)	Proposed embryological support for evolution
		(v)	Theory of biogenesis

- (1) a(ii), b(iv), c(i), d(iii)
- (2) a(i), b(v), c(iv), d(iii)
- (3) a(iii), b(v), c(iv), d(i)
- (4) a(i), b(iii), c(ii), d(iv)

55. Intestinal perforation and death may occur in severe cases of

- (1) Typhoid fever
- (2) Pneumonia
- (3) Ringworm
- (4) Malaria

56. Read the following statements.

- (a) Hind brain of frogs consists of cerebellum, pons and medulla oblongata.
- (b) About 7% of CO_2 is carried in a dissolved state through plasma.
- (c) In female cockroaches, the 7th sternum is boat shaped.
- (d) The electrical activity of the heart can be recorded from the body surface by using electrocardiogram and the graphical representation of recording is called electrocardiograph.

Choose the option representing **correct** statements only.

- (1) (a) and (b)
- (2) (b) and (c)
- (3) (c) and (d)
- (4) (a) and (d)

57. Genetic engineering is better over traditional hybridization in

- (1) Introducing desirable genes only
- (2) Introducing undesirable genes
- (3) Introducing both desirable and undesirable genes
- (4) Being easy to perform

58. The red and white muscle fibres can be distinguished from each other on the basis of all of the following, **except**

- (1) Amount of myoglobin
- (2) Dependence on aerobic process for energy
- (3) Amount of sarcoplasmic reticulum
- (4) Contractile ability due to actin and myosin filaments

59. All of the following can suppress sperm motility and the fertilising capacity of sperms and can also be used as emergency contraceptives, **except**

- (1) Cervical caps
- (2) Cu-T
- (3) Cu-7
- (4) Multiload-375

60. Four-chambered heart is found in

- (1) Humans only
- (2) Mammals only
- (3) All vertebrates only
- (4) Some reptiles, birds and mammals

61. Read the given statements.

Statement I : Health, for a long time by early Greeks, was considered as a state of body and mind where there was a balance of certain 'humors'.

Statement II : Both neural system and endocrine system help in maintaining our health by influencing immune system.

Select the **correct** option.

- (1) Only statement I is correct
- (2) Only statement II is correct
- (3) Both statements I and II are correct
- (4) Neither statement I nor statement II is correct

62. Which of the following groups of hormones influence the maintenance of water and electrolytes balance in human body?

- (1) TCT, ADH, cortisol
- (2) Thyroxine, aldosterone, ADH
- (3) ADH, glucagon, thymosin
- (4) Insulin, cortisol, thymosin

63. Toxicity testing on non-transgenic as well as transgenic mice was performed for a certain drug. Which of the following is correct w.r.t this toxicity testing?

- (1) Both of them will give results at the approximately same time.
- (2) The non-transgenic mice will give results in lesser time compared to the transgenic mice.

The transgenic mice will give results in lesser time as compared to the non-transgenic mice as the former is less sensitive to the toxic substance being studied.

- (3) compared to the non-transgenic mice as the former is less sensitive to the toxic substance being studied.
- (4) toxic substances, hence, they will give results faster than the non-transgenic mice.

64. Which of the following is **incorrect** w.r.t. notochord?

- (1) Ectodermally derived
- (2) Rod-like structure
- (3) Present on the dorsal side in some animals
- (4) Absent in animals ranging from phylum Porifera to Echinodermata

65. If we ligate foreign DNA at *Bam* HI site of pBR322, then the resultant recombinants will show

- (a) Resistance to ampicillin
- (b) Sensitivity to tetracycline
- (c) Resistance to tetracycline
- (d) Sensitivity to ampicillin

Choose the correct option.

- (1) (a) and (b)
- (2) (a) and (c)
- (3) (b) and (d)
- (4) (c) and (d)

66. Females are sterile as ovaries are rudimentary besides lack of other secondary sexual characters in:

- (1) Down's syndrome
- (2) Klinefelter's syndrome
- (3) Turner's syndrome
- (4) Cystic fibrosis

67. 'X-body' of Henking was in fact a chromosome and that is why it was given the name _____.
Select the correct option to fill in the blank.

- (1) X-chromosomes
- (2) Y-chromosomes
- (3) Allosomes
- (4) Polysome

68. When the *Escherichia coli* growing in a medium containing $^{35}\text{SO}_4$ are infected with T₂ bacteriophage, which of the following results would be observed?

- (1) Radioactivity will be observed in the supernatant
- (2) The bacteriophage formed in the bacteria will not show radioactivity
- (3) Radioactivity will be seen in the genetic material of the bacteria
- (4) Radioactivity will be seen in the capsid of the newly formed bacteriophage

69. A transcription unit is 7000 nucleotides long. This unit may use _____ nucleotides to make a polypeptide that can have _____ amino acids.

Select the **correct** option to fill in the blanks A and B.

	A	B
(1)	3500	1750
(2)	7000	2330
(3)	2100	650
(4)	1400	500

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

70. The structural genes in *lac* operon is responsible for all, **except**

- (1) Synthesis of repressor protein
- (2) Formation of product that increases the permeability of the cell to β -galactosides
- (3) Synthesis of enzyme that transfers acetyl group to β -galactosides
- (4) Synthesis of enzyme that hydrolyses lactose

71. Choose the **wrong** match

(1)	F. Meischer	–	First identified DNA in nucleus
(2)	Wilkins & Franklin	–	X-ray diffraction data of DNA
(3)	Altman	–	Termed nuclein to DNA
(4)	E. Chargaff	–	Gave ratios between nitrogenous bases A and T and G and C for dsDNA

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

72. Which of the following is a ribozyme in bacteria for the formation of peptide bond?

- (1) 28 S rRNA
- (2) 5.8 S rRNA
- (3) 23 S rRNA
- (4) 16 S rRNA

73. The blind approach of simply sequencing the whole set of genome that contained all the coding and non-coding sequence, and later assigning different regions in the sequence with functions is termed as

- (1) Bioinformatics
- (2) Sequence annotation
- (3) EST approach
- (4) Transcriptome method

74. Water samples are collected from various sections of STPs and following table is created:

Sample	BOD (mg/L)
A	20
B	5
C	300
D	400

(River water is considered to be least polluted) What will be the source of sample 'C'?

- (1) River water
- (2) Primary effluent
- (3) Secondary effluent
- (4) Untreated sewage water

75. Which of the following cyanobacterium can fix the atmospheric nitrogen in an aquatic environment?
- (1) *Rhizobium*
 - (2) *Clostridium*
 - (3) *Oscillatoria*
 - (4) *Azotobacter*
76. You never see any cattle or goats browsing on *Calotropis* because
- (1) They produce highly poisonous cardiac glycosides
 - (2) They have spines instead of leaves
 - (3) They are distasteful to their herbivores
 - (4) They are too long in heights to reach by herbivores
77. An overwhelming majority of animals and nearly all plants cannot maintain a constant internal environment. They are
- (1) Regulators
 - (2) Conformers
 - (3) Partial regulators
 - (4) Suspenders
78. Producers in the following food chain converts the energy from sun into the biomass of 100000 J energy. What would be amount of energy that will pass from 2nd to 3rd trophic level?
- Plants → Caterpillar → Sparrow → Hawk
- (1) 1000 J
 - (2) 100 J
 - (3) 10 J
 - (4) 10000 J
79. $\log S = \log C + Z \log A$
Value of Z or regression coefficient in above equation for relatively smaller area is generally
- (1) Between 1 to 2
 - (2) 1 to 1.2
 - (3) Between 0.1 to 0.2
 - (4) 0.5 to 1
80. Read the following statements and choose the option which is true for them.
- Statement A:** A stable community should not show too much variation in productivity from year to year.
- Statement B:** A stable community must be resistant to invasions by alien species.
- (1) Both the statements are true
 - (2) Both the statements are false
 - (3) Only statement A is true
 - (4) Only statement B is true
81. Nematode-specific genes were introduced into the tobacco plant by using
- (1) Disarmed Ti-plasmid of *Agrobacterium*
 - (2) *Meloidogyne incognita*
 - (3) *Bacillus thuringiensis*
 - (4) Retrovirus
82. Choose the **odd** one w.r.t. ends produced by *EcoRI*.
- (1) Sticky ends
 - (2) Cohesive ends
 - (3) Overhanging stretches on each strand of DNA
 - (4) Blunt ends
83. Select the **mismatch** w.r.t. average composition of cells.
- (1) Protein – 10-15%
 - (2) Lipids – 7%
 - (3) Nucleic acids – 5-7%
 - (4) Water – 70-90%
84. Read the following statements and select the correct option w.r.t implantation.
- Statement A:** During implantation, blastocyst gets embedded in the endometrium of uterus.
- Statement B:** Uterine cells divide rapidly and cover the blastocyst once it gets attached to the uterine wall.
- (1) Both statements A and B are correct
 - (2) Only statement B is correct
 - (3) Both statements A and B are incorrect
 - (4) Only statement A is correct
85. In *EcoRI*, the letter R is derived from
- (1) Genus
 - (2) Species
 - (3) Strain RY 13
 - (4) Strain Rd
86. The proteins encoded by which of the following genes control corn borer?
- (1) *cryIAc*
 - (2) *cryIIAb*
 - (3) *cryIAb*
 - (4) *rop*
87. Choose the **odd** one w.r.t. contraceptive methods.
- (1) MTPs
 - (2) Withdrawal method
 - (3) Implants
 - (4) IUDs

88. In a standard ECG, P-wave represents

- (1) Depolarisation of ventricles
- (2) Repolarisation of auricles
- (3) Depolarisation of atria
- (4) Excitation of auricles and ventricles

89. Which of the following are **correct** w.r.t. hymen?

- (a) Is often torn during first intercourse
 - (b) Partially covers vaginal opening
 - (c) It forms the part of female external genitalia
- Select the correct option.

- (1) (a), (b) only
- (2) (b), (c) only
- (3) (a), (c) only
- (4) (a), (b), (c)

90. Which of the following hormones is secreted by placenta only?

- (1) Oxytocin
- (2) Estrogen
- (3) Progesterone
- (4) hPL

CHEMISTRY

91. The element with highest negative electron gain enthalpy belongs to

- (1) Period 2, group 18
- (2) Period 2, group 17
- (3) Period 3, group 17
- (4) Period 2, group 1

92. Given below are the two statements.

Statement I: A polyatomic molecule can have zero dipole moment.

Statement II: It is necessary for a polyatomic molecule to have linear geometry to have dipole moment zero
In light of above statements, choose the correct answer from the options given below.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are correct
- (4) Both statement I and statement II are incorrect

93. Match List-I with List-II.

	List-I (Molecule)		List-II (Shape)
a.	ClF ₃	(i)	Bent
b.	SO ₂	(ii)	See-saw
c.	SF ₄	(iii)	Trigonal Pyramidal
d.	NF ₃	(iv)	T-shape

The correct match is

- (1) a(ii), b(iii), c(i), d(iv)
- (2) a(iv), b(i), c(iii), d(ii)
- (3) a(iv), b(i), c(ii), d(iii)
- (4) a(i), b(iv), c(ii), d(iii)

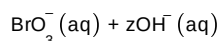
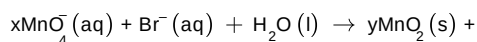
94. Which of the following will make acidic buffer?

- (1) 50 mL of 0.1 M CH₃COOH + 50 mL of 0.1 M NaOH
- (2) 50 mL of 0.1 M HCl + 100 mL of 0.1 M NaOH
- (3) 100 mL of 0.1 M CH₃COONa + 50 mL of 0.1 M HCl
- (4) 50 mL of 0.1 M NaOH + 25 mL of 0.1 M CH₃COOH

95. The correct order of solubility for the given salts CuCl ($K_{sp} = 1.7 \times 10^{-6}$), PbCO₃ ($K_{sp} = 7.4 \times 10^{-14}$) and SnS ($K_{sp} = 1.0 \times 10^{-25}$) is

- (1) CuCl > SnS > PbCO₃
- (2) CuCl > PbCO₃ > SnS
- (3) PbCO₃ > SnS > CuCl
- (4) SnS > PbCO₃ > CuCl

96. In the chemical reaction,



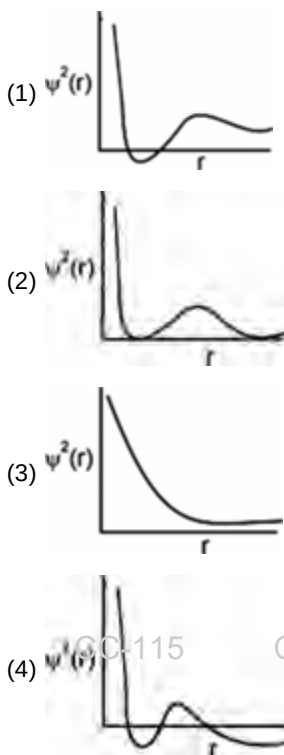
x, y and z respectively are

- (1) 2, 2, 1
- (2) 1, 1, 2
- (3) 2, 2, 2
- (4) 1, 1, 4

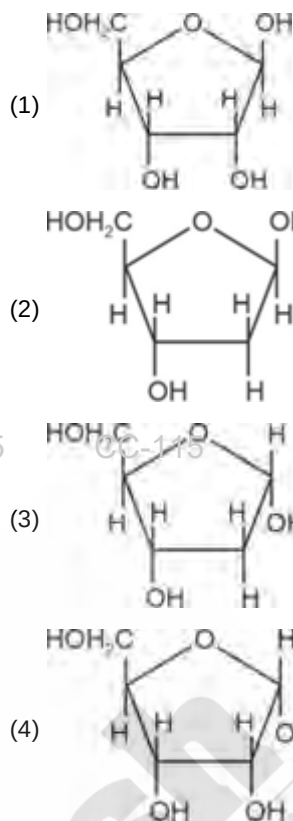
97. Which of the following reactions is an example of disproportionation reaction?

- (1) $2\text{PbNO}_3(\text{s}) \rightarrow 2\text{PbO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
- (2) $\text{NaH}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$
- (3) $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaOCl}(\text{aq}) + \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- (4) $\text{CH}_4(\text{g}) + 4\text{Cl}_2(\text{g}) \rightarrow \text{CCl}_4(\text{l}) + 4\text{HCl}(\text{g})$

98. The correct probability density distribution curve with respect to distance r for 2s orbital is



99. Structure of the sugar moiety present in DNA is



100. The number of photons emitted from a monochromatic light of wavelength 5000 Å to give 1 J energy is ($h = 6.63 \times 10^{-34} \text{ Js}$)

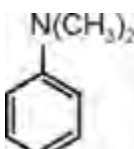
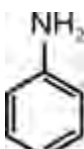
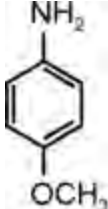
- (1) 2.5×10^{16}
- (2) 3.5×10^{18}
- (3) 2.5×10^{18}
- (4) 2.5×10^{19}

101. Consider the following statements

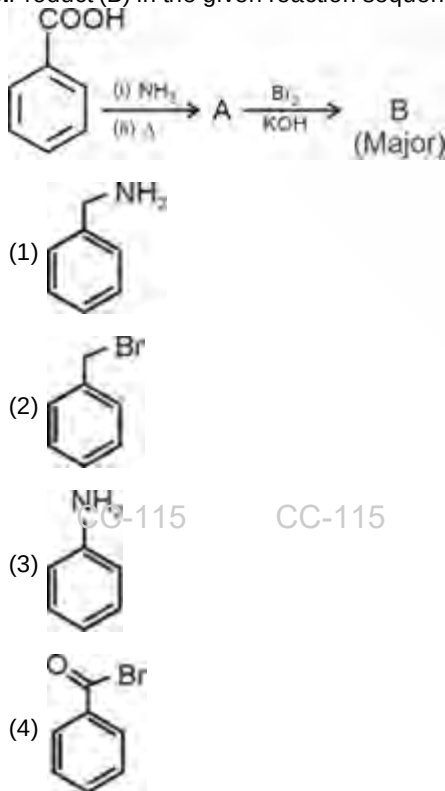
- (a) Sucrose is dextrorotatory compound
 - (b) Maltose is composed of two α -D-glucose units
 - (c) Number of -OH groups present in lactose are 10
- The **correct** statement(s) is/are

- (1) (a) and (b) only
- (2) (b) only
- (3) (b) and (c) only
- (4) (a), (b) and (c)

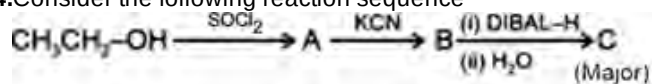
102. Which among the following compounds has highest pK_b ?

- (1) 
- (2) 
- (3) 
- (4) 

103. Product (B) in the given reaction sequence is



104. Consider the following reaction sequence



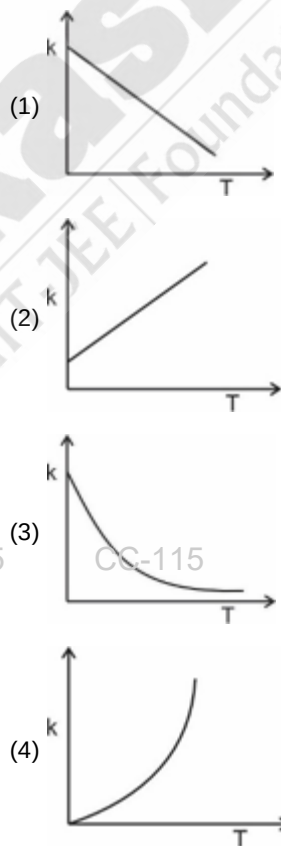
Incorrect statement about C is

- (1) It reduces Fehling's solution
 (2) It does not give haloform reaction
 (3) It gives positive test with 2,4 DNP
 (4) It undergoes disproportionation reaction in alkaline medium

105. For the first order reaction, $\text{A(g)} \rightarrow \text{B(g)} + 2\text{C(g)}$, if P_i is the initial pressure of A and P_t is the total pressure at time t , then the rate constant (k) of the reaction is given by

- (1) $k = \frac{2.303}{t} \log \left(\frac{3P_i}{2P_i - P_t} \right)$
 (2) $k = \frac{2.303}{t} \log \left(\frac{P_i}{2P_i - P_t} \right)$
 (3) $k = \frac{2.303}{t} \log \left(\frac{P_i}{3P_i - P_t} \right)$
 (4) $k = \frac{2.303}{t} \log \left(\frac{2P_i}{3P_i - P_t} \right)$

106. The correct plot for Arrhenius equation showing variation of rate constant (k) with temperature (T) is



107. Given below are the two statements.

Statement I: Ethanol on reaction with copper at 573 K gives ethanoic acid as major product.

Statement II: Ethanol in presence of sulphuric acid at 413 K gives ethoxyethane as major product.

In light of above statements, choose the correct answer.

- (1) Statement I is incorrect but statement II is correct
- (2) Statement I is correct but statement II is incorrect
- (3) Both statement I and statement II are correct
- (4) Both statement I and statement II are incorrect

108. The azeotropic composition of aqueous nitric acid is

- (1) 68% HNO_3 + 32% H_2O by mass
- (2) 76% HNO_3 + 24% H_2O by mass
- (3) 52% HNO_3 + 48% H_2O by mass
- (4) 45% HNO_3 + 55% H_2O by mass

109. If air contains O_2 and N_2 in 1 : 4 mole ratio, then number of moles of N_2 dissolved in two moles of water at 5 atm pressure and 25°C temperature will be (Given : K_H for N_2 in water at 25°C is 10^5 atm).

- (1) 4×10^{-5}
- (2) 2×10^{-5}
- (3) 6×10^{-8}
- (4) 8×10^{-5}

110. An element has two isotopes having atomic mass 10 u and 15 u respectively. If the % abundance of lighter isotope is 80% then the average atomic mass of the element is

- (1) 9 u
- (2) 11 u
- (3) 12 u
- (4) 14 u

111. Identify the incorrect match.

	Multiple	Prefix
(1)	10^{-15}	Femto
(2)	10^{-12}	Pico
(3)	10^{15}	Tera
(4)	10^2	Hecto

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

112. Calculate the enthalpy of formation of HCl if bond dissociation enthalpies of H_2 , Cl_2 and HCl are 434, 242 and 431 kJ mol^{-1} respectively

- (1) 93 kJ mol^{-1}
- (2) -93 kJ mol^{-1}
- (3) 186 kJ mol^{-1}
- (4) -186 kJ mol^{-1}

113. If equilibrium constant for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ is K_C then the equilibrium constant for the reaction $\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g})$ is

- (1) K_C
- (2) $\frac{1}{K_C}$
- (3) $\frac{1}{\sqrt{K_C}}$
- (4) $\sqrt{K_C}$

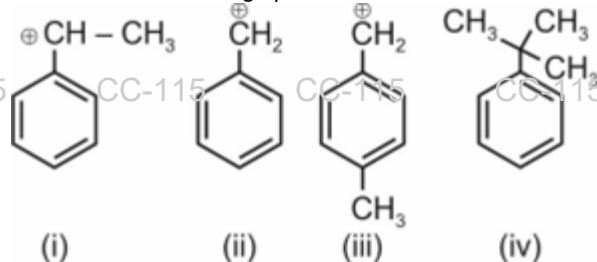
114. If the solubility product of $\text{Ni}(\text{OH})_2$ is 2×10^{-15} , then its molar solubility in aqueous solution of 0.2 M NaOH will be

- (1) $5 \times 10^{-15} \text{ M}$
- (2) $5 \times 10^{-14} \text{ M}$
- (3) $2 \times 10^{-3} \text{ M}$
- (4) $5 \times 10^{-13} \text{ M}$

115. Incorrect about Borax is

- (1) Aqueous solution of Borax is alkaline in nature
- (2) On heating Borax in Bunsen burner flame with cobalt oxide, blue colour bead is formed
- (3) Heating of Borax produces sodium metaborates
- (4) It contains four 'B-O-B' bonds

116. Consider the following species



Hyperconjugation will be observed in

- (1) (i) and (ii) only
- (2) (ii) and (iii) only
- (3) (i) and (iii) only
- (4) (i), (ii), (iii) and (iv)

117. In sulphur estimation 0.25 g of an organic compound gave 0.466 g of barium sulphate. Percentage of sulphur in the compound is (Molecular mass of $\text{BaSO}_4 = 233 \text{ u}$)

- (1) 45.5%
- (2) 16.4%
- (3) 56.5%
- (4) 25.6%

118. Match List-I with List-II.

	List-I (Compounds)		List-II (Number of σ bonds)
a.	Phenanthrene	(i)	18
b.	Gammexane	(ii)	16
c.	Naphthalene	(iii)	19
d.	Neopentane	(iv)	26

The correct match is

- (1) a(iv), b(i), c(iii), d(ii)
- (2) a(iii), b(i), c(iv), d(ii)
- (3) a(i), b(iv), c(iii), d(ii)
- (4) a(ii), b(iii), c(i), d(iv)

119. Consider the following statements.

- (a) Ethane is obtained as product on heating sodium acetate with sodalime.
- (b) Solvent used in Wurtz reaction is dry ether.
- (c) Propyne on reaction with dihydrogen in presence of finely divided palladium catalyst gives propane as major product.

The **correct** statements are

- (1) (a) & (b) only
- (2) (b) & (c) only
- (3) (a) & (c) only
- (4) (a), (b) & (c)

120. Identify the incorrect statement.

- (1) Solubility of a gas in liquid decreases with increase of temperature
- (2) At high altitudes, the partial pressure of oxygen is higher than that at the ground level
- (3) The tanks used by scuba divers are filled with air diluted with helium.
- (4) To increase the solubility of CO_2 in soft drinks and soda water, the bottles are sealed under high pressure

121. The emf of cell

$\text{Ni(s)}|\text{Ni}^{2+}(0.16 \text{ M})||\text{Ag}^+(0.002 \text{ M})|\text{Ag(s)}$,
at 298 K is ($E^\circ_{\text{cell}} = 1.05 \text{ V}$)

- (1) -0.91 V
- (2) $+0.46 \text{ V}$
- (3) $+0.91 \text{ V}$
- (4) -0.75 V

122. Select the correct order against the indicated property

- (1) Boiling point : $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (2) Acidic nature : $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (3) Dissociation enthalpy : $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (4) Melting point : $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$

123. Given below are the two statements one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Ionisation of an atom is an endothermic process.

Reason (R): The order of first ionization enthalpy of given elements follows the order $\text{O} < \text{N} < \text{F} < \text{Ne}$.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (4) (A) is false but (R) is true

124. Consider the following statements.

- a. Nitrous acid can disproportionate in acid solution into nitric acid and nitrous oxide
 - b. Single $\text{N} - \text{N}$ bond is weaker than the single $\text{P} - \text{P}$ bond
 - c. Very pure nitrogen can be obtained by the thermal decomposition of Barium azide
 - d. Liquid dinitrogen is used as refrigerant to preserve biological materials
- Correct statements are

- (1) a and c only
- (2) a and d only
- (3) b, c and d only
- (4) c and d only

125. The correct order of ionic radii of Eu^{3+} , La^{3+} and Yb^{3+} is

- (1) $\text{Eu}^{3+} > \text{La}^{3+} > \text{Yb}^{3+}$
- (2) $\text{Yb}^{3+} > \text{Eu}^{3+} > \text{La}^{3+}$
- (3) $\text{La}^{3+} > \text{Eu}^{3+} > \text{Yb}^{3+}$
- (4) $\text{La}^{3+} > \text{Yb}^{3+} > \text{Eu}^{3+}$

126. Given below are the two statements.

Statement I: Rhombic sulphur is insoluble in water but is readily soluble in CS_2 .

Statement II: Both rhombic and monoclinic sulphur have S_8 molecules.

In light of above statements, choose the correct answer.

- (1) Statement I is correct but statement II is incorrect
- (2) Both statement I and statement II are correct
- (3) Both statement I and statement II are incorrect
- (4) Statement I is incorrect but statement II is correct

127. Select the incorrect statement regarding metal carbonyls among the following

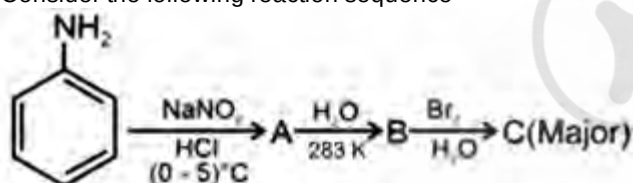
- (1) Metal-carbon bond possesses both σ and π character
 $M - C$ σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of the metal
- (2) $M - C$ π -bond is formed by the donation of a pair of electrons from a filled d orbital of metal into vacant bonding π -orbital of CO
- (3) The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal
- (4) The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal

128. Consider the following statements

- (a) If a compound rotates plane polarized light in clockwise direction then it is called dextrorotatory
 - (b) A molecule which is non-superimposable on its mirror image is chiral and optically active
 - (c) Propan-2-ol is an achiral molecule
- The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a), (b) and (c)
- (4) (a) and (c) only

129. Consider the following reaction sequence



Major product (C) is

- (1)
- (2)
- (3)
- (4)

130. Polar protic solvent among the following is

- (1) Acetone
- (2) DMSO
- (3) Ethanol
- (4) DMF

131. Most acidic compound among the following is

- (1)
- (2)
- (3)
- (4)

132. Match List-I with List-II

List-I Reaction	List-II Name
a.	(i) Gatterman reaction Koch
b.	(ii) Kolbe's reaction
c.	(iii) Hell-Volhard-Zelinsky reaction
d.	(iv) Rosenmund reduction

The correct match is

- (1) a(iv), b(i), c(ii), d(iii)
- (2) a(iv), b(i), c(iii), d(ii)
- (3) a(i), b(iv), c(ii), d(iii)
- (4) a(ii), b(iii), c(iv), d(i)

133. Given below are the two statements

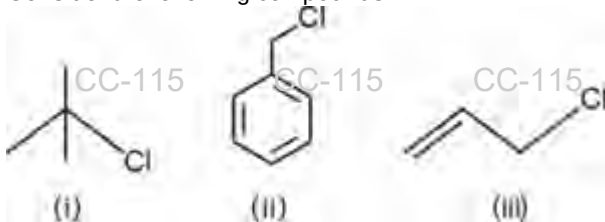
Statement I: SO_2 is a colourless gas and is highly soluble in water.

Statement II: SO_2 reacts with chlorine in the presence of charcoal to give thionyl chloride.

In light of above statements, choose the correct answer.

- (1) Statement I is correct but statement II is incorrect.
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are correct.
- (4) Both statement I and statement II are incorrect.

134. Consider the following compounds



The correct order of $\text{S}_{\text{N}}1$ rate for the given compounds is

- (1) (ii) > (i) > (iii)
- (2) (iii) > (i) > (ii)
- (3) (i) > (iii) > (ii)
- (4) (i) > (ii) > (iii)

135. Match List-I with List-II.

	List-I Complex		List-II Number of unpaired electrons
a.	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	Zero
b.	$[\text{Co}(\text{H}_2\text{O})_6]^{3+}$	(ii)	5
c.	$[\text{FeF}_6]^{3-}$	(iii)	2
d.	$[\text{NiCl}_4]^{2-}$	(iv)	1

The correct match is

- (1) a(i), b(iv), c(ii), d(iii)
- (2) a(iv), b(i), c(iii), d(ii)
- (3) a(ii), b(iii), c(iv), d(i)
- (4) a(iv), b(i), c(ii), d(iii)

136. Sum of the numbers 436.32 g, 227.2 g and 0.301 g is

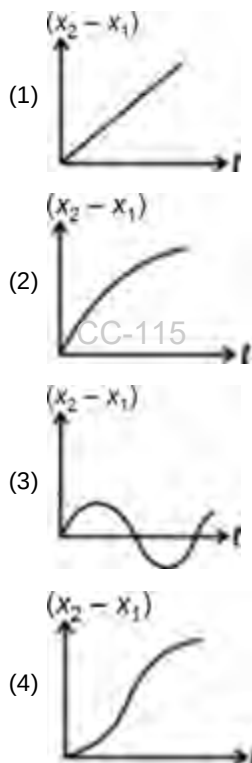
- (1) 663.821 g
- (2) 663.8 g
- (3) 6663.82 g
- (4) 664 g

137. With a constant acceleration along x-axis, a car covers a distance of 20 m during fourth second of its motion and 25 m during fifth second of its motion. Distance covered by it during third second of motion was

- (1) 15 m
- (2) 18 m
- (3) 30 m
- (4) 10 m

PHYSICS

138. Two stones are thrown up from the edge of a cliff 300 m high with initial speed of 10 m/s and 20 m/s. Which of the following graph best represents the variation of relative position of second stone with respect to first stone till both the stones are in air? (neglect air resistance)



139. In the following question, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**.

Assertion (A) : If two particles, moving along straight line with constant velocities have to meet, the relative velocity must be along the line joining the two particles.

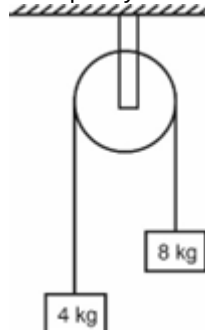
Reason (R) : Relative motion means motion of one particle as viewed from the other particle.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true and the (R) is not the correct explanation of the (A)
- (3) (A) is true statement but (R) is false
- (4) Both (A) & (R) are false statements

140. Equation of trajectory of a projectile is $y = \sqrt{3}x - 5x^2$. Then angle of projection with vertical is (Assume x-axis as horizontal and y-axis as vertical)

- (1) 45°
- (2) 30°
- (3) 60°
- (4) 53°

141. Two blocks of masses 4 kg and 8 kg are connected to fixed ideal pulley as shown and system is released from rest.



Acceleration of 4 kg block after release is

- (1) $\frac{2g}{3}$ upwards
- (2) $\frac{g}{3}$ upwards
- (3) $\frac{2g}{3}$ downwards
- (4) $\frac{g}{3}$ downwards

142. A simple pendulum hanging freely stayed at rest in vertical, because in this position

- (1) Potential energy is maximum
- (2) Kinetic energy is minimum
- (3) Potential energy is minimum
- (4) Net force acting is towards point of suspension

143. State the incorrect statement among the following

When conservative force acts on a system and performs

- (1) positive work on body, potential energy of body decreases
- (2) In an inelastic collision of two bodies linear momentum of system remains conserved
- (3) In case of inelastic collision, final kinetic energy is less than initial kinetic energy of the system
- (4) Total kinetic energy of system is conserved no matter what internal and external forces on body are present

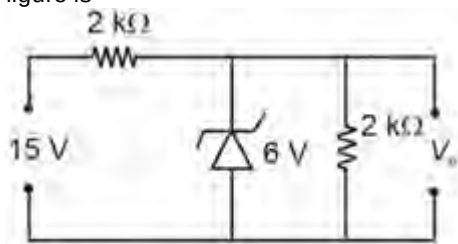
144. In the following questions, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**

Assertion (A) : The photo-diode can be used as a photodetector to detect optical signals.

Reason (R) : It is easier to observe the change in the current with change in the light intensity, if a forward bias is applied to photodiode.

- (1) If both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) If both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) If (A) is true statement but (R) is false
- (4) If both (A) & (R) are false statements

145. The value of output voltage V_0 in the circuit as shown in the figure is



- (1) 9 V
(2) 4.5 V
(3) 12 V
(4) 6 V
146. Consider the following statements:
(a) Nuclear density is directly proportional to cube root of mass number.
(b) Binding energy per nucleon is maximum for elements having mass number $A > 170$.
(c) When two deuterium nuclei fuse together to form a tritium nucleus we get a proton
(d) Neutrons and protons are bound in a nucleus by the short range weak nuclear force.
The correct statement(s) is/are
- (1) Both (a) and (b)
(2) Only (b)
(3) Only (c)
(4) Both (c) and (d)
147. A nucleus of mass number 238 at rest emits an alpha particle with kinetic energy 5.4 MeV. The Q value of reaction is
- (1) 5.49 MeV
(2) 5.88 MeV
(3) 5.67 MeV
(4) 5.60 MeV
148. An electron in a hydrogen atom makes a transition from $n = n_1$ to $n = n_2$. The time period of revolution of the electron in the initial state is eight times that in final state. The possible value of n_1 and n_2 are
- (1) $n_1 = 4, n_2 = 2$
(2) $n_1 = 8, n_2 = 2$
(3) $n_1 = 8, n_2 = 1$
(4) $n_1 = 6, n_2 = 2$

149. Two particles of masses 2 kg and 6 kg located at the point (1 m, 1 m, 1 m) and (2 m, 2 m, 1 m) respectively. The distance of centre of mass from 2 kg mass will be

- (1) $\frac{\sqrt{2}}{4}$ m
(2) $\frac{3\sqrt{2}}{4}$ m
(3) $\frac{1}{\sqrt{2}}$ m
(4) $\frac{3\sqrt{2}}{8}$ m

150. Three equal masses of 3 kg each are fixed at the vertices of an equilateral triangle ABC. The gravitational force acting on mass 2 kg placed at the centroid of triangle is

- (1) Zero
(2) 6.67×10^{-3} N
(3) 9×10^{-9} N
(4) Data is insufficient

151. A rocket is fired vertically with a speed half of the escape speed from the earth's surface. How far from the earth does the rocket go before returning to the earth? [Given mean radius of earth is R]

- (1) $R/3$ from earth's centre
(2) $4R/3$ from the earth's surface
(3) $4R/3$ from the earth's centre
(4) $7R/3$ from the earth's surface

152. In the following question, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**.

Assertion (A): On increasing intensity of light, more number of photoelectrons are emitted.

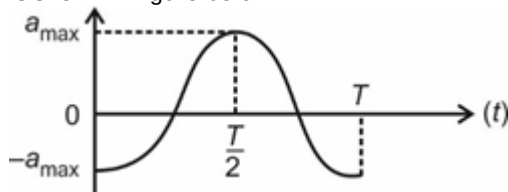
Reason (R): The number of electrons emitted does not depend on intensity of incident radiations

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
(2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
(3) (A) is true statement but (R) is false
(4) Both (A) & (R) are false statements

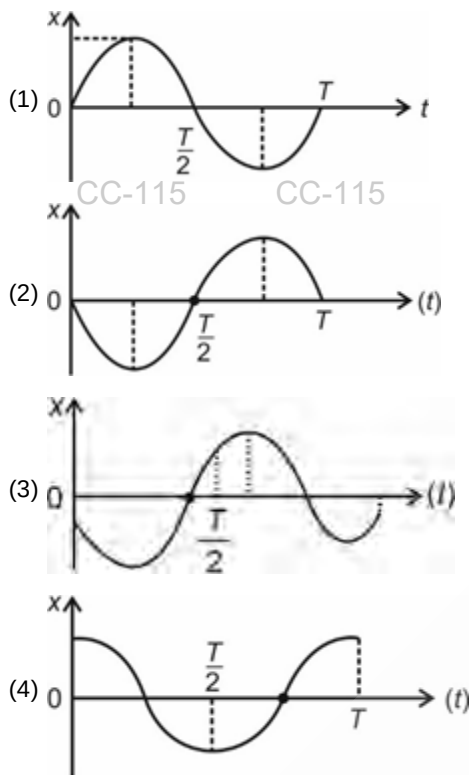
153. In an electron microscope electron is accelerated by 150 kV then de-Broglie wavelength is λ . If voltage is increased by 300% then the de Broglie wavelength of electron will be

- (1) $\lambda/2$
(2) λ
(3) 2λ
(4) $\lambda/4$

- 154.** Acceleration versus time graph for a particle executing SHM is shown in figure below.



Corresponding position-time graph will be



- 155.** If a plane wave front is incident on a convex lens then the emerging wavefront will be

- (1) Plane
- (2) Spherical
- (3) Parabolic
- (4) Hyperbolic

- 156.** A progressive wave of frequency 500 Hz is travelling with a speed of 330 m/s in air. The distance between the two points which have a phase difference 30° is

- (1) 0.11 m
- (2) 0.055 m
- (3) 0.22 m
- (4) 0.025 m

- 157.** Two closed pipes produce 10 beats per second when emitting their fundamental modes. If their lengths are in ratio of 25 : 26. Then their fundamental frequencies in Hz are

- (1) 270, 280
- (2) 250, 270
- (3) 260, 250
- (4) 260, 280

- 158.** For the damped oscillator, if time taken for its amplitude of vibrations to drop to half of its initial value is T then time taken for amplitude to drop to one eighth amplitude is

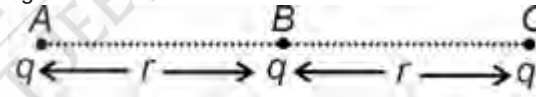
- (1) $t = 2T$
- (2) $t = 3T$
- (3) $t = 4T$
- (4) $t = T/2$

- 159.** Consider the following statements:

- (a) Capacitance of any capacitor also depends upon the charge of capacitor
 - (b) When capacitors are in series then charge on each capacitor will have same value.
 - (c) When we connect the two charged capacitors then the total electrostatic energy will be always conserved.
- The correct statement(s) is/are

- (1) Only (a)
- (2) Only (b)
- (3) Only (c)
- (4) Both (a) and (c)

- 160.** Minimum work done by external agent in assembling three identical charges from infinity to given location as shown in figure is



- (1) $\frac{5}{8\epsilon_0} \frac{q^2}{r}$
- (2) $\left(\frac{5}{8\pi\epsilon_0} \frac{q^2}{r} \right)$
- (3) $\frac{5}{2\pi\epsilon_0} \frac{q^2}{r}$
- (4) $\frac{3q^2}{8\pi\epsilon_0 r}$

- 161.** A system of two charges $q_A = 2.5 \times 10^{-7} \text{ C}$ and $q_B = -2.5 \times 10^{-7} \text{ C}$ are located at points A: (0, 0, -15 cm) and B: (0, 0, 15 cm) respectively. The electric dipole moment of system is

- (1) $2.5 \times 10^{-7} \text{ C m}$
- (2) $5 \times 10^{-7} \text{ C m}$
- (3) $7.5 \times 10^{-8} \text{ C m}$
- (4) Zero

162. Experimental observations show that for given solid material, the magnitude of strain produced is same whether the stress is tensile or compressive. The ratio of tensile stress to longitudinal strain is defined as Young's modulus and is denoted by $Y = \sigma/\epsilon$. The length of a metal wire is l_A when the tension in it is T_A and is l_B when tension is T_B . The natural length of wire is

- (1) $\frac{T_B l_B + T_A l_A}{T_A + T_B}$
- (2) $\frac{T_B l_B - T_A l_A}{T_A - T_B}$
- (3) $\frac{T_B l_A - T_A l_B}{T_B - T_A}$
- (4) $\frac{T_B l_B + T_A l_A}{T_A - T_B}$

163. A U-tube contains water and methylated spirit separated by mercury. The mercury columns in the two arms are in level with 12 cm of water in one arm and 15 cm of spirit in other. The specific gravity of spirit is

- (1) 0.8
- (2) 0.4
- (3) 0.5
- (4) 1

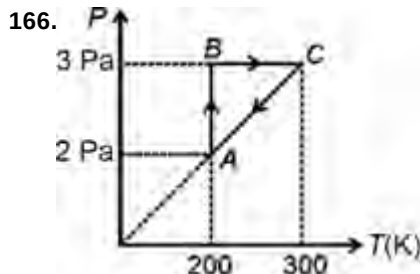
164. In thermodynamic processes, correct match of column-I with column-II is

	Column-I Type of process		Column-II Feature
a.	Isothermal	(i)	Volume constant
b.	Isobaric	(ii)	Pressure constant
c.	Isochoric	(iii)	No heat flow between system and surroundings
d.	Adiabatic	(iv)	Temperature constant

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iv), b(i), c(iii), d(ii)
- (3) a(iv), b(ii), c(iii), d(i)
- (4) a(iv), b(ii), c(i), d(iii)

165. For an ideal gas, total energy is equally distributed in all possible energy modes, with each mode has an average energy equal to $\frac{1}{2} K_B T$, and each vibrational mode has energy contribution of

- (1) $\frac{1}{3} K_B T$
- (2) $K_B T$
- (3) $\frac{3}{2} K_B T$
- (4) $\frac{1}{4} K_B T$



166.

Consider a $P - T$ diagram for a cyclic process. Given, $U_A = 200$ J and $U_C = 300$ J.

What will be the value of U_B ?

- (1) 250 J
- (2) 200 J
- (3) 300 J
- (4) 150 J

167. Estimate the average drift speed of conduction electrons in a conductor of cross-sectional area 10^{-7} m^2 carrying current of 1.5 A. The number density of conduction electrons is $8.5 \times 10^{28} \text{ m}^{-3}$.

- (1) 2.2 mm s^{-1}
- (2) 1.1 mm s^{-1}
- (3) 3.3 mm s^{-1}
- (4) 0.1 mm s^{-1}

168. A maximum current can be drawn from a cell when

- (1) Cell is open
- (2) Cell is connected with a finite external resistance
- (3) External resistance is equal to internal resistance
- (4) Cell is short circuited

169. Consider the following statements:

Statement (A): All magnetism is due to circulating currents i.e. there cannot be magnetic field without current.

Statement (B): The orbital magnetic moment of an electron in the ground state of a hydrogen atom is equal to one Bohr Magnetron.

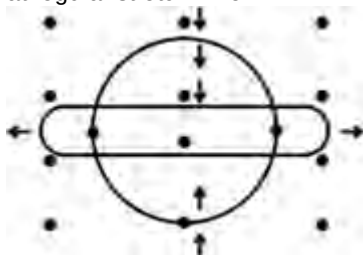
Choose the correct option.

- (1) Statement (A) is correct while statement (B) is incorrect.
- (2) Statement (B) is correct while statement (A) is incorrect.
- (3) Both the statements are correct
- (4) Both the statements are incorrect

170. A galvanometer coil has a resistance of 12Ω and the meter show full scale deflection for a current of 1 mA. The resistance required to be added to convert the galvanometer into a voltmeter of range 0 to 10 V will be

- (1) 9988Ω in parallel
- (2) 9988Ω in series
- (3) 10000Ω in parallel
- (4) 10000Ω in series

171. A circular coil is being deformed into a narrow straight wire at regular stretch. Then



- (1) The direction of induced current is clockwise
 - (2) The direction of induced current is anticlockwise
 - (3) Magnetic flux through coil increases
 - (4) The amount of charge flowing in coil depends on time
172. Two circular coils one of small radius r and other of larger radius R ($r \ll R$) are placed co-axially with centres coinciding. The mutual inductance of the arrangement is

- (1) $\frac{\mu_0 \pi R^2}{r}$
- (2) $\frac{\mu_0 \pi r^2}{2R}$
- (3) $\frac{\mu_0 \pi r R}{(r + R)}$
- (4) $\frac{2\mu_0 \pi r^2}{R}$

173. Consider a series LCR circuit in which reactance and resistance are 100Ω each. When the circuit is connected to ac source 220 V, 50 Hz, then current drawn from the source is

- (1) $2.2\sqrt{2}$ A
- (2) $1.1\sqrt{2}$ A
- (3) $3.3\sqrt{2}$ A
- (4) 2.2 A

174. 15 gm of ice at 0°C is mixed with 300 gm of water at 50°C in a container. There is no heat loss due to radiation and water equivalent of container is ignored. What will be final temperature of water?

- (1) 5.3°C
- (2) 6.7°C
- (3) 12.3°C
- (4) 43.8°C

175. The relation between coefficient of linear expansion (α) and coefficient of volume expansion (γ) for solids is

- (1) $\gamma = 2\alpha$
- (2) $3\gamma = \alpha$
- (3) $\gamma = 3\alpha$
- (4) $2\gamma = 3\alpha$

176. Which of the following conclusion can be drawn if an electromagnetic wave satisfies the Ampere's law for vacuum?

- (1) $\mu_0 \epsilon_0 = \frac{B_0}{E_0 c}$
- (2) $c = \frac{B_0}{E_0}$
- (3) $\oint \vec{E} \cdot d\vec{l} = \frac{-d\phi_B}{dt}$
- (4) Both (1) and (2)

177. If a container of height 17.3 cm is filled with a liquid of refractive index μ . The bottom of container appear to be raised by 3.46 cm when seen from above. Refractive index of liquid is

- (1) 1.33
- (2) 1.25
- (3) 1.52
- (4) 1.1

178. The refracting angle and minimum angle of deviation of a prism is same as 60° . The refractive index of prism is

- (1) 1.732
- (2) 1.414
- (3) 2
- (4) 2.82

179. Match the elements of List-I with List-II

	List-I		List-II
(A)	Simple microscope	(E)	Image magnified, inverted and virtual
(B)	Compound microscope	(F)	Image virtual, erect and high resolution
(C)	Astronomical telescope	(G)	Virtual, inverted and high resolution
(D)	Terrestrial telescope	(H)	Image virtual, erect and enlarged

- (1) A-H, B-F, C-E, D-G
- (2) A-H, B-E, C-G, D-F
- (3) A-H, B-E, C-F, D-G
- (4) A-F, B-G, C-E, D-G

180. In a resonance tube at room temperature two successive resonance lengths of air column are 25 cm and 80 cm. If the frequency of tuning fork is 340 Hz then the speed of sound at that temperature is

- (1) 354 m/s
- (2) 340 m/s
- (3) 374 m/s
- (4) 350 m/s